

MOHAMED MUSTAFA | Steel Structures Design

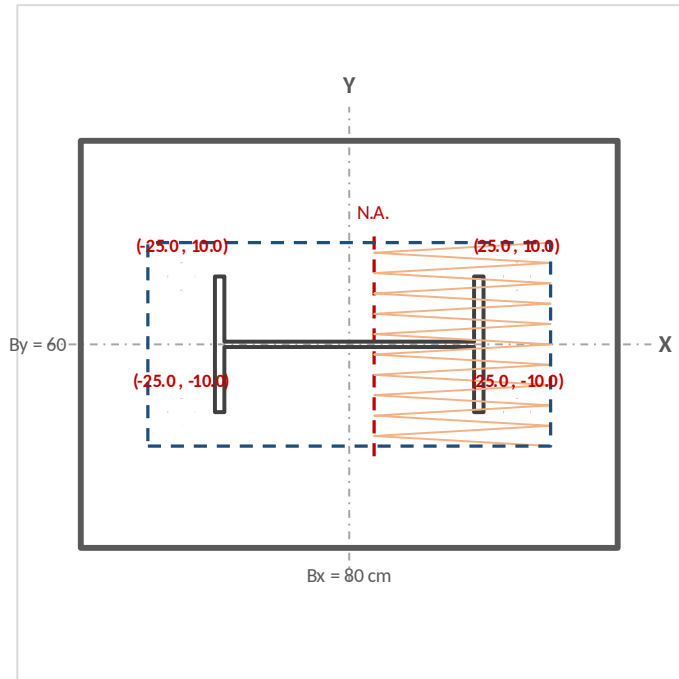
Column Base Plate & Anchor Bolt Forces - ECP 205 (ASD / LRFD)

DESIGN BY : **ENG. Mohamed Mustafa**

Project ID. :

Data Entry

<u>Design Method :</u>	ASD	
<u>Forces Input :</u>	Column	
<u>Column Section :</u>	I-Section	
<u>Moment About :</u>	Major	
<u>Section Depth h =</u>	400	mm
<u>Section Width b =</u>	200	mm
<u>Flange tf =</u>	14	mm
<u>Web tw =</u>	8	mm
<u>Plate Steel :</u>	ST 37	
<u>Base Plate Lx =</u>	60	cm
<u>Base Plate Ly =</u>	30	cm
<u>Plate Thickness =</u>	4	cm
<u>Weld Type :</u>	Fillet	
<u>Weld Size s =</u>	12	mm
<u>Concrete Fcu =</u>	250	kg/cm2
<u>Pedestal Bx =</u>	80	cm
<u>Pedestal By =</u>	60	cm
<u>Bolt Ø (sheet 2) =</u>	30	mm
<u>Base Plate :</u>	Safe	



Plan to scale - updates with the inputs:

- ⊕ bolt (X , Y) in cm
- pedestal - - - base plate
- column profile
- - - neutral axis (red)
- //// compression zone under the plate

Column Forces (service) :

<u>Axial Force N =</u>	20	ton
<u>N is :</u>	Compression	
<u>Moment M =</u>	8	t.m
<u>Shear Vx =</u>	3	ton
<u>Shear Vy =</u>	1	ton

(+) M compresses the +X side (Major) / +Y side (Minor)

Anchor Bolts Layout :

IDEA only - not used

Bolt No.	X (cm)	Y (cm)	T (ton)	Vx (ton)	Vy (ton)
1	-25.0	-10.0			
2	25.0	-10.0			
3	25.0	10.0			
4	-25.0	10.0			
5					
6					
7					
8					
9					
10					
11					
12					

X along h, Y along b, from the column centre (= pedestal centre). Leave unused rows empty. T, Vx, Vy only when Forces Input = IDEA.

Data Output:

Plate & Bolt Properties:

Fy =	2.4	t/cm2	Fu =	3.6	t/cm2	
Bolt As =	5.51	cm2	n = Es/Ec =	9.5		As = 0.78·π·Ø²/4 ; Ec = 4400√fcu
L (along M) =	60.0	cm	B =	30.0	cm	
A = L × B =	1800	cm2	Z = B·L²/6 =	18000	cm3	

Load Case:

e =	40.0	cm	L/6 =	10.0	cm
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Compression with $e > L/6 \rightarrow$ partial bearing + tension in the bolts.

Bearing & Neutral Axis:

d i = bolt to compressed edge

c =	26.3	cm	k =	2.711	kg/cm3
q max =	71.3	kg/cm2	q min =	0.0	kg/cm2

Anchor Bolt Forces:

Ti = k · n · As · (d i - c)							
Vi = √(Vx² + Vy²) / n (equal share)							
Bolt No.	T (ton)	b (cm)	a (cm)	w (cm)	Q (ton)	T+Q (ton)	V (ton)
1	4.06	5.0	5.0	15.0	0.39	4.45	0.79
2	0.00	5.0	5.0	15.0	0.00	0.00	0.79
3	0.00	5.0	5.0	15.0	0.00	0.00	0.79
4	4.06	5.0	5.0	15.0	0.39	4.45	0.79

$$\text{Prying (ECP 205): } Q = T \cdot \left[\frac{1}{2} - \frac{w \cdot t^4}{(30 \cdot a \cdot b^2 \cdot As)} \right] / \left[(3a/4b + 1) \cdot (a/3b + 1) + \frac{w \cdot t^4}{(6 \cdot a \cdot b^2 \cdot As)} \right]$$

b = bolt to column face, a = bolt to plate edge $\leq 1.25 b$, w = plate width per bolt, t = plate thickness. Highlighted row = critical bolt.

TO SHEET 2: max (T + Q) = 4.45 ton (bolt 1) | max V = 0.79 ton

Concrete Bearing Check:

Allowable Bearing =	70.0	kg/cm2	√(A2/A1) =	1.33	
Actual Bearing =	71.3	kg/cm2	< 93.3	Safe	SR = 76%

Base Plate Thickness Check:

Allowable Stress =	1.73	t/cm2	σ = 0.72 Fy (ECP 205)		
Cantilever m =	11.0	cm	n =	7.0	cm from 0.95h / 0.8b
M (bearing) =	3.71	t.cm/cm	M (bolts) =	1.22	t.cm/cm

bearing: cantilever m (pressure diagram) and n (pressure at the column face) | bolts: max(T·b - Q·a , Q·a) / w

Design Thickness =	3.59	cm	Used = 4 cm	Safe
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Column Weld Check:

Weld Length Lw =	152.8	cm	Sw = lw/c =	1893	cm2
Throat a = 0.7 s =	0.84	cm	Allowable =	0.72	t/cm2 0.2 Fu
Actual Stress =	0.66	t/cm2	< 0.72	Safe	line method, all around

Notes: design aid only - the responsible engineer must verify all results. See sheet "Notes".

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Anchor Bolt & Washer Plate Design - ECP 205 (ASD / LRFD)

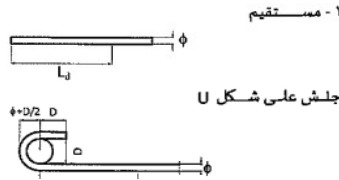
DESIGN BY : **ENG. Mohamed Mustafa**

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Data Entry

Design Method :	ASD
Bolt Properties :	ST 52
Force in bolt =	4.45 ton
Embedment hef =	120 cm
Bolt Diameter =	30 mm
Concrete Fcu =	250 kg/cm ²
Consider Bond :	No
Washer Diameter =	8 cm
Washer Properties :	ST 37
Selected Thickness =	1.6 cm

Anchor Shape : **Straight**



Anchor type: **Plain**

Safe

Data Output:

Bolt Properties:

Fy =	3.6 t/cm ²	Fu =	5.2 t/cm ²
Bolt Area =	7.07 cm ²	Perimeter =	9.42 cm
Bond Surface Area =	1131.0 cm ²		
Bolt Allowable tension =	9.46	> 4.45	Safe

Concrete Properties:

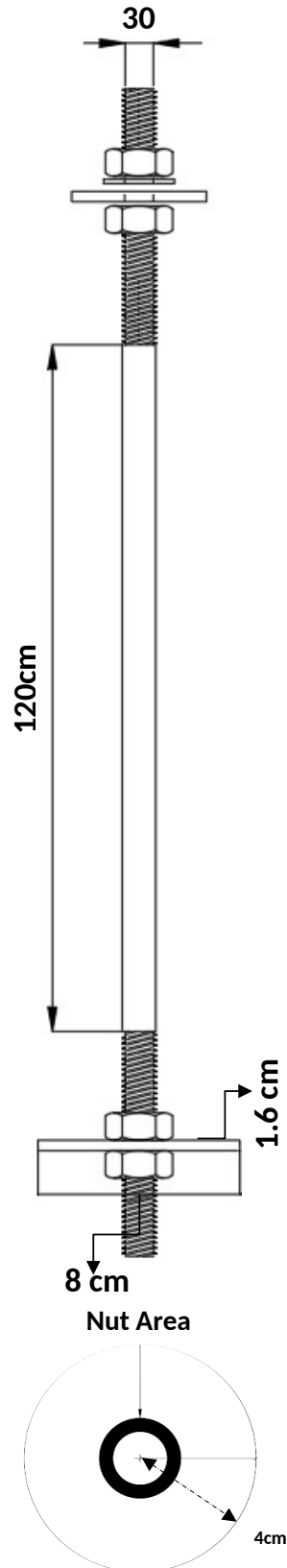
Allowable Bearing =	70.0 kg/cm ²	C25
Concrete bond =	8.25 kg/cm ²	ASD
Bearing x $\sqrt{(A_2/A_1)}$ =	2.00	Allow = 140.0 kg/cm ²

Washer Properties:

Fy =	2.4 t/cm ²	Fu =	3.6 t/cm ²
Area =	43.20 cm ²	Bearing =	2.16 t/cm ²
Force transfer by bolt =	0 kg		0.00 ton
Force in Washer =	4450 kg		4.45 ton
Max Moment In Washer plate =	0.68 t.cm		

Washer Plate design:

Allowable Stress =	1.73 t/cm ²
Washer Plate Thickness Check:	$t = \sqrt{((6M) / (b \times \sigma))}$
Design Thickness =	1.54 cm
Concrete Bearing Check:	
Actual Bearing =	103.0 kg/cm ² < 140.0 SR=74% Safe
Washer Bearing Check:	
Allowable Stress =	2.16 t/cm ²
Actual Bearing =	0.50 < 2.16 t/cm Use Nut diameter = 4.5 cm Safe
Washer Pull Check:	
Allowable Stress =	0.84 t/cm ²
Actual Stress =	0.30 < 0.84 t/cm ² Safe
Bolt Shear Check:	$(f_t / F_t)^2 + (q / q_b)^2 \leq 1.0$
Bolt Allowable shear =	7.17 > 0.79 Safe
Interaction =	0.23 < 1.0 Safe
Washer Place Check:	$D/2 + 5 \leq c \text{ min} ; D \leq s \text{ min}$
c min =	15.0 cm
s min =	20.0 cm Safe



Concrete breakout, side-face blowout, pryout and the pedestal reinforcement → sheet "3-Pedestal".

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Concrete Pedestal Design - ACI 318-19 Ch.17 (Anchorage) + ECP 203 (Section)

DESIGN BY : **ENG. Mohamed Mustafa**

Project ID. :

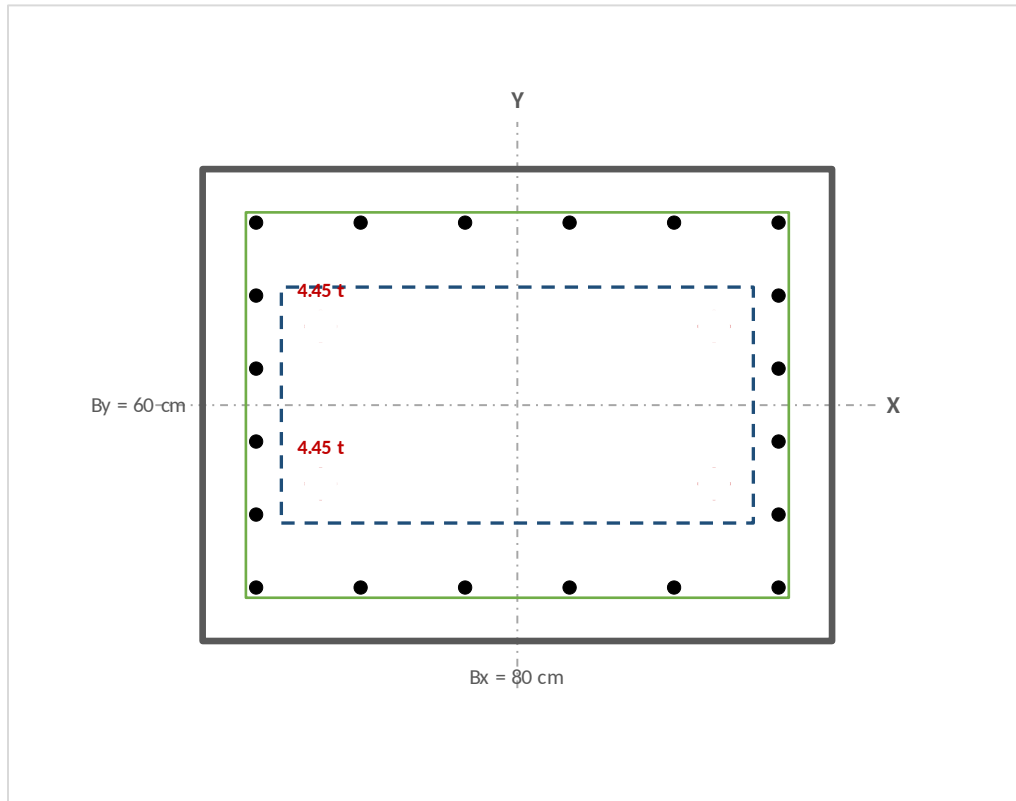
Data Entry

Pedestal :	Reinforced	Rebar Steel :	36/52
Pedestal Height =	150	Main Bar Ø =	16 mm
Clear Cover =	5	Tie Ø =	10 mm
		Status :	Safe

Data From Sheets 1 & 2:

Bx × By =	80 × 60	cm	f'c = 0.8f _{cu} =	19.6	MPa
hef =	120	cm	Bolt Ø =	30	mm
Washer D =	8	cm	A _{brg} =	43.2	cm ²
N =	20.00	ton (C)	M =	8.00	t.m
V _x =	3.00	ton	V _y =	1.00	ton
Tension bolts =	2		Σ(T+Q) =	8.90	ton
Max (T+Q) =	4.45	ton	Design =	ASD	

Pedestal Plan (to scale) :



⊕ bolt (label = T + Q in tension) ● main bars ■ ties (green) - - - base plate ■ pedestal

Data Output:

ASD: anchorage strength = $\phi \cdot N_n / 1.5$ compared with service forces; pedestal section with forces $\times 1.5$ (ECP 203 ultimate).

Breakout in Tension:

			$\phi N_{cbg} = \phi \cdot (A_{Nc} / A_{Nco}) \cdot \psi_{ec} \cdot \psi_{ed} \cdot \psi_c \cdot N_b$		
hef' =	43.3	cm	Nb =	399.4	kN $N_b = 10 \cdot \sqrt{f'_c} \cdot \text{hef}'^{1.5}$
ANc =	4800	cm ²	ANco =	16900	cm ² $A_{Nco} = 9 \cdot \text{hef}'^2$
ψ_{ed} =	0.77		ψ_{ec} =	1.00	$0.7 + 0.3 \cdot c / 1.5 \text{hef}' ; 1 / (1 + e'N / 1.5 \text{hef}')$
Capacity =	43.6	kN		< 87.3 kN	By Rft. $\phi = 0.75$ (with rft) ; $\Sigma(T+Q)$ of bolts in tension

Side-Face Blowout:

			$N_{sb} = 13 \cdot ca_1 \cdot \sqrt{A_{brg}} \cdot \sqrt{f'_c} \cdot (1 + ca_2 / ca_1) / 4$		
ca1 =	15.0	cm	ca2 =	20.0	cm only when hef > 2.5 · ca1
Capacity =	165.5	kN		> 43.7 kN	Safe

Breakout in Shear:

			$V_b = \min[0.6 \cdot (l_e/d_a)^{0.2} \cdot \sqrt{d_a}, 3.7] \cdot \sqrt{f'_c} \cdot ca_1^{1.5}$				
<u>Direction</u>	<u>ca1 (cm)</u>	<u>Avc/Avco</u>	<u>ψed</u>	<u>Vb (kN)</u>	<u>φVcbg (kN)</u>	<u>V (kN)</u>	<u>Check</u>
Shear X	15.0	1.33	0.97	30.1	23.3	29.4	By Rft.
Shear Y	20.0	1.33	0.85	46.3	31.5	9.8	Safe

$\psi_c, V = 1.2$ (edge bars), $\phi = 0.75$; $\psi_h = \sqrt{(1.5 \cdot ca_1 / H)} \geq 1$; $l_e = \min(\text{hef}, 8d_a)$; all shear on the front row

Pryout:

			$\phi V_{cp} = \phi \cdot k_{cp} \cdot N_{cp} ; k_{cp} = 2, \phi = 0.70$		
Ncp =	152.8	kN	ΣV =	31.0	kN N_{cp} = breakout of all bolts
Capacity =	142.7	kN		> 31.0 kN	Safe

Tension - Shear Interaction:

			$N_{ua} / \phi N_n + V_{ua} / \phi V_n \leq 1.2$ (ACI §17.8)		
$N_{ua} / \phi N_n$ =	0.47		$V_{ua} / \phi V_n$ =	0.35	governing mode (steel/concrete/rft)
Sum =	0.82			< 1.2	Safe

Pedestal Section (N + M):

			$M_u = M_u + N_u \cdot (d - h/2) ; A_s = (0.447 \cdot f_{cu} \cdot b \cdot a - N_u) / (f_y / 1.15)$		
Nu =	30.00	ton	Mu base =	18.75	t.m $\times 1.5 ; M_u = M + V \cdot H$
h × b =	80 × 60	cm	d =	73.2	cm h along the moment
As req / face =	3.49	cm ²	As min =	38.4	cm ² 0.8% · Bx · By
Mr (strain comp.) =	52.15	t.m		> 18.75	Safe $\epsilon_{cu} = 0.003, 0.447 f_{cu}, f_y / 1.15$
Axial capacity =	517.0	ton		> 30.0	Safe $0.35 f_{cu} \cdot A_c + 0.67 f_y \cdot A_s$
Shear qu =	1.08	kg/cm ²		< 9.9	Safe $q_{cu} = 0.24 \cdot \sqrt{f_{cu} / 1.5}$

Reinforcement Details:

			$A_s \text{ anchor} = T_u / (0.75 \cdot f_y)$ (ACI §17.5.2.1)		
Tu max =	6.67	ton	As / bolt =	2.47	cm ² $T_u = 1.5 \times (T + Q)$
Bars / bolt =	3 Ø16	around each tension bolt (within 0.5 hef)			
Main bars =	20 Ø16	(6 on each face ⊥ M + 4 on each side)	As =	40.2	cm ²
Dev. length ld =	61	cm	Bar L =	185	cm hef + ld

$l_d = f_y \cdot d_b / (2.1 \cdot \sqrt{f'_c})$ ($\emptyset \leq 20, 1.7$ for larger) ≥ 30 cm - bars hooked at the top, developed below the bolt head

Top ties = 3 Ø10 closed ties around the bolts, top 12.5 cm (shear $A_s = 1.67$ cm²)
Body ties = Ø10 @ 15 cm (≤ 15 cm, 12 Ø)

top ties: $A_s = V_u / (0.75 \cdot f_y)$, 2 legs per tie, at least 3 Ø10 / 2 Ø12 within the top 12.5 cm (ACI §17.7.2.5, §10.7.6.1.6)

REINFORCEMENT: 20 Ø16 (L ≥ 185 cm) | 3 Ø16 around each tension bolt | top ties 3 Ø10 | ties Ø10 @ 15 cm

Notes: design aid only - the responsible engineer must verify all results. See sheet "Notes".

Column Base Plate, Anchor Bolt & Pedestal Design - by Eng. Mohamed Mustafa

Steel Structures Design Engineer | <https://mohamedmostafa20-max.github.io/>

HOW TO USE

Fill the white boxes only - every other cell is calculated and protected. Work from sheet 1 to sheet 3.

1-Base Plate: design method, column section (I / Box / Pipe), base plate, weld, concrete, pedestal size, the column forces N, M (one axis), Vx, Vy and the X / Y of every bolt from the column centre (up to 12 bolts). The sheet finds the force in every bolt (elastic neutral axis with bolt stiffness $n \cdot A_s$), adds the prying force, and checks concrete bearing, base plate thickness and the column weld.

Forces Input = IDEA: type T, Vx, Vy of every bolt from IDEA StatiCa (biaxial moments, special cases). Prying is then taken as already included, and plate / weld / bearing are left to IDEA.

2-Anchor Bolt: bolt grade, diameter, embedment, anchor shape, bond option and washer steel. The force is the largest (T + Q) from sheet 1, the shear is the largest bolt shear. Washer diameter and thickness are selected automatically.

3-Pedestal: plain / reinforced, pedestal height, cover and bar sizes. All concrete anchorage checks and the pedestal reinforcement are produced, with the summary in the green line at the end.

METHOD & REFERENCES

- Bolt forces: rigid plate, linear strain. $e \leq L/6$ full bearing ($q = N/A \pm M/Z$); tension with the whole plate lifting: bolt-group formula; otherwise the neutral axis c is solved from $N = k[Bc^2/2 - nA_s \sum (d_i - c)]$ and $M = k[Bc^2/2(L/2 - c/3) + nA_s \sum (d_i - c)(d_i - L/2)]$, $n = E_s/E_c$, $E_c = 4400\sqrt{f_{cu}}$.
- Prying (ECP 205): $Q = T[1/2 - wt^4/(30ab^2A_s)] / [(3a/4b + 1)(a/3b + 1) + wt^4/(6ab^2A_s)]$, $a \leq 1.25b$.
- Base plate: $t = \sqrt{(6M / (b \cdot \sigma))}$, $\sigma = 0.72F_y$ (ASD) or $1.35F_y$ (LRFD, $\phi M_p = 0.9F_y t^2/4$). Bearing side: cantilever m from $0.95h$ (0.8b minor axis, 0.8D pipe) with the actual pressure diagram, and n with the pressure at the column face. Bolt side: $\max(T \cdot b - Q \cdot a, Q \cdot a)$ per width w.
- Concrete bearing under the plate: ASD $70 / 110 \text{ kg/cm}^2$ ($F_{cu} 250 / 350$), LRFD $0.67f_{cu}/1.5$, $\times \sqrt{(A_2/A_1)} \leq 2$.
- Column weld: line method around the profile, $f = \sqrt{((N/L_w + M/S_w)^2 + (V/L_w)^2)}/a \leq 0.2F_u$ (ASD) / $0.45F_u$ (LRFD).
- Bolt steel (ECP 205): tension $0.33F_u$ (ASD) / $0.75F_u$ (LRFD), shear $0.25F_u / 0.30F_u$, on $A_s = 0.78A_g$; interaction $(f_t/F_t)^2 + (q/q_b)^2 \leq 1$.
- Washer plate, bond, nut bearing and washer punching: as the original sheet (ECP 203 / ECP 205 / AISC J7).
- Concrete anchorage: ACI 318-19 Ch.17 in SI units, $f'_c = 0.8f_{cu}$, cracked concrete. ASD allowable = $\phi N_n / 1.5$. Breakout in tension of the bolts in tension (with $\psi_{ec,N}$ and the h_{ef} rule for 3+ near edges), side-face blowout, breakout in shear (both directions, all shear on the front row), pryout ($k_{cp} = 2$), interaction (17.8).
- Reinforced pedestal: anchor reinforcement $A_s = T_u/(0.75f_y)$ per bolt (17.5.2.1) when the breakout is not enough; top ties for shear $A_s = V_u/(0.75f_y)$ ($\geq 3\emptyset 10 / 2\emptyset 12$, 10.7.6.1.6). Both are increased when needed so that the tension-shear interaction (17.8) stays ≤ 1.2 .
- Pedestal section (ECP 203): $M_u = M + V \cdot H$ at the pedestal base, ASD loads $\times 1.5$. Required steel by the big-eccentricity method, then the provided bars are checked by strain compatibility ($\epsilon_{cu} = 0.003$, $0.447f_{cu}$, $f_y/1.15$, all bars). Minimum steel 0.8% of $B_x \cdot B_y$, bar spacing $\leq 25 \text{ cm}$.
- Shear in the anchor bolts is shared equally by all bolts (Column input).

DISCLAIMER

This workbook is a design aid. The responsible engineer must review every input, assumption and result and confirm them against the governing code before use. No warranty is given for any use of this workbook.

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